

SARDAR PATEL UNIVERSITY
B.C.A (3rd SEM) - EXAMINATION OCT 2025
US03CBCA52 – DATA STRUCTURES - I

**Date: 07-10-2025(Tuesday)****Time: 02:00 PM to 05:00 PM****NOTE: Right hand figure indicates the marks of each question.****Total Marks: [70]****Q. 1 Multiple Choice Questions (One marks each)****[10]**

- 1 _____ are the building blocks of program.
(A) Information (B) Data Structure
(C) Data (D) Data Types
- 2 An array is a _____ data structure.
(A) Composite (B) Non-composite
(C) Unordered (D) Heterogeneous
- 3 Index is also known as _____.
(A) Subscript (B) Non-subscript
(C) Base (D) Type
- 4 The term "push" and "pop" is related to _____.
(A) Array (B) Stack
(C) Queue (D) All of these
- 5 In the _____ notation the operators are written after the operands.
(A) Prefix (B) Postfix
(C) Infix (D) None of these
- 6 An operation that is used to delete an element from a stack is known as _____.
(A) Push (B) Pop
(C) Peep (D) Change
- 7 A Queue is logically a _____ type of list.
(A) First-in First-out (B) First-in Last-out
(C) Last-in First-out (D) Last-in Last-out
- 8 The line formed for paying fees for college is an example of _____.
(A) Stack (B) Simple Queue
(C) Circular Queue (D) Priority Queue
- 9 A _____ is a collection of information items about a particular entity.
(A) Record (B) Database
(C) Entity (D) None of these
- 10 A record item that uniquely identifies a record in a file is called a _____.
(A) Key (B) Item
(C) Sequence Key (D) None of these

Q. 2 Short Questions. (Attempt any ten)

[20]

- 1 Define Data Structure.
- 2 List out advantages of Data Structure.
- 3 Differentiate Primitive data structure and Non-Primitive Data Structure.
- 4 List operations carried out on a Stack. Define ANY ONE of them.
- 5 Explain prefix notation.
- 6 What is Push and Pop operations of a stack?
- 7 What are Ascending Priority Queues? Explain.
- 8 Write a note on Simple Queues.
- 9 Define queue. List the types of Queues.
- 10 Define the terms: Record and File.
- 11 What do you mean by file organization? Why it is Required?
- 12 Define the terms: Multigraph and Cycle.

Q. 3 A] Explain in brief Classification of Data Structure.

[05]

B] List out operations used on Data Structure. Explain any three.

[05]

OR

Q. 3 A] Define array. Explain 1-D array with declaration and initialization.

[05]

B] Explain the representation of 1-D array in the memory.

[05]

Q. 4 A] Define stack. And also explain notations in detail.

[05]

B] Write an algorithm to insert an element into a Stack.

[05]

OR

Q. 4 A] Explain Infix to Postfix conversion.

[05]

B] Explain recursion in detail with example.

[05]

Q. 5 A] Explain the Priority Queues with example.

[05]

B] Write an algorithm to Insert an element in a Simple queue.

[05]

OR

Q. 5 A] Explain the Circular Queues with an example.

[05]

B] Write an algorithm to Delete an element from a Simple queue.

[05]

Q. 6 A] Explain in detail the structure of Sequential File.

[05]

B] Write a short note on Multiple buffering.

[05]

OR

Q. 6 A] Define: (1) Database, (2) Key, (3) Graph, (4) Diagraph, (5) Cycle.

[05]

B] Write a short note on Single buffering.

[05]

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